

Foundations of Robotics

Homework 3

1. Which of the following matrices are rotation matrices? Explain why (not).

$$R_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$R_3 = \begin{bmatrix} 0.75 & 0.25 & 0.6124 \\ 0.25 & 0.75 & -0.6124 \\ -0.6124 & 0.6124 & 0.5 \end{bmatrix} \quad R_4 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_5 = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad R_6 = \begin{bmatrix} 0.5 & 0 & 0.5 \\ 0.5 & -0.5 & 0 \\ 0 & -0.5 & 0.5 \end{bmatrix}$$

2. (Exercise 3.1) Please solve the exercise introduced as follows.

Exercise 3.1 In terms of the $\hat{x}_s, \hat{y}_s, \hat{z}_s$ coordinates of a fixed space frame $\{s\}$, the frame $\{a\}$ has its $\hat{x}_a$ -axis pointing in the direction $(0, 0, 1)$ and its $\hat{y}_a$ -axis pointing in the direction $(-1, 0, 0)$, and the frame $\{b\}$ has its $\hat{x}_b$ -axis pointing in the direction $(1, 0, 0)$ and its $\hat{y}_b$ -axis pointing in the direction $(0, 0, -1)$.

- (a) Draw by hand the three frames, at different locations so that they are easy to see.
- (b) Write down the rotation matrices R_{sa} and R_{sb} .
- (c) Given R_{sb} , how do you calculate R_{sb}^{-1} without using a matrix inverse? Write down R_{sb}^{-1} and verify its correctness using your drawing.
- (d) Given R_{sa} and R_{sb} , how do you calculate R_{ab} (again without using matrix inverses)? Compute the answer and verify its correctness using your drawing.
- (e) Let $R = R_{sb}$ be considered as a transformation operator consisting of a rotation about $\hat{x}$ by -90° . Calculate $R_1 = R_{sa}R$, and think of R_{sa} as a representation of an orientation, R as a rotation of R_{sa} , and R_1 as the new orientation after the rotation has been performed. Does the new orientation R_1 correspond to a rotation of R_{sa} by -90° about the world-fixed $\hat{x}_s$ -axis or about the body-fixed $\hat{x}_a$ -axis? Now calculate $R_2 = RR_{sa}$. Does the new orientation R_2 correspond to a rotation of R_{sa} by -90° about the world-fixed $\hat{x}_s$ -axis or about the body-fixed $\hat{x}_a$ -axis?

- (f) Use R_{sb} to change the representation of the point $p_b = (1, 2, 3)$ (which is in $\{b\}$ coordinates) to $\{s\}$ coordinates.
- (g) Choose a point p represented by $p_s = (1, 2, 3)$ in $\{s\}$ coordinates. Calculate $p' = R_{sb}p_s$ and $p'' = R_{sb}^T p_s$. For each operation, should the result be interpreted as changing coordinates (from the $\{s\}$ frame to $\{b\}$) without moving the point p or as moving the location of the point without changing the reference frame of the representation?
- (h) An angular velocity w is represented in $\{s\}$ as $\omega_s = (3, 2, 1)$. What is its representation ω_a in $\{a\}$?
- (i) By hand, calculate the matrix logarithm $[\hat{\omega}]\theta$ of R_{sa} . (You may verify your answer with software.) Extract the unit angular velocity $\hat{\omega}$ and rotation amount θ . Redraw the fixed frame $\{s\}$ and in it draw $\hat{\omega}$.
- (j) Calculate the matrix exponential corresponding to the exponential coordinates of rotation $\hat{\omega}\theta = (1, 2, 0)$. Draw the corresponding frame relative to $\{s\}$, as well as the rotation axis $\hat{\omega}$.

3. (Exercise 3.15) Please solve the exercise discussed as follows.

Exercise 3.15

- (a) Given a fixed frame $\{0\}$ and a moving frame $\{1\}$ initially aligned with $\{0\}$, perform the following sequence of rotations on $\{1\}$:
 1. Rotate $\{1\}$ about the $\{0\}$ frame $\hat{x}$ -axis by α ; call this new frame $\{2\}$.
 2. Rotate $\{2\}$ about the $\{0\}$ frame $\hat{y}$ -axis by β ; call this new frame $\{3\}$.
 3. Rotate $\{3\}$ about the $\{0\}$ frame $\hat{z}$ -axis by γ ; call this new frame $\{4\}$.

What is the final orientation R_{04} ?

- (b) Suppose that the third step above is replaced by the following: “Rotate $\{3\}$ about the $\hat{z}$ -axis of frame $\{3\}$ by γ ; call this new frame $\{4\}$.” What is the final orientation R_{04} ?

4. (Exercise 3.9) Exploiting the known properties of rotation matrices, determine the minimum number of arithmetic operations (multiplication and division, addition and subtraction) required to multiply two rotation matrices.

5. (Exercise 3.28) Please solve the exercise discussed as follows.

Exercise 3.28 Assume that the space-frame angular velocity is $\omega_s = (1, 2, 3)$ for a moving body with frame $\{b\}$ at

$$R = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

relative to the space frame $\{s\}$. Calculate the body's angular velocity ω_b in $\{b\}$.

6. (Exercise 3.10) This is a bonus question and is not required in this homework. However, you are encouraged to think over about the following question, because it is really useful in practice.

Exercise 3.10 Because arithmetic precision is only finite, the numerically obtained product of two rotation matrices is not necessarily a rotation matrix; that is, the resulting rotation A may not exactly satisfy $A^T A = I$ as desired. Devise an iterative numerical procedure that takes an arbitrary matrix $A \in \mathbb{R}^{3 \times 3}$ and produces a matrix $R \in SO(3)$ that minimizes

$$\|A - R\|^2 = \text{tr} (A - R)(A - R)^T.$$

(Hint: See Appendix D for the relevant background on optimization.)